Spring

The Spring Self-Evolving Potential Function Trainer:
A Training-Is-Audit Framework for
Molecular Dynamics Potential Learning

SCX

202671

v1.0 | | SCX —

Abstract

Spring—— NEPDeepMDACE Spring $M > 1$ Yajie LyapunovΨ_t M_t Cercis Spring$\Psi_t \to 0 t \to \infty$1 $M_t \Sigma_0(\mathcal{D})$-2 Yajie$M_t$3 SpringNEP$M = 1$DeepMD$M = 1ACEM = 1$ Spring""

YajieLyapunovCercisSpring

Contents

1

1.1

Molecular Dynamics, MD$V(\mathbf{r}_1, \ldots, \mathbf{r}_N)$

Lennard-JonesEAMReaxFF

Machine Learning Potentials, MLPs Behler-Parrinello2007

- **NEPNeuroevolution Potential, 2022** $M = 1$
- **DeepMDDeep Potential Molecular Dynamics, 2018** -$M = 1$
- **ACEAtomic Cluster Expansion, 2019** $M = 1$
- **GAPGaussian Approximation Potential, 2010** $M = 1$
- **MTPMoment Tensor Potential, 2016** $M = 1$

———— / ——””””

1.2

SCX [?, ?] $\mathcal{D} = \{(\mathbf{R}_i, E_i, \mathbf{F}_i)\}_{i=1}^{N_{\text{data}}}$ $\mathbf{R}_i E_i \mathbf{F}_i$ $V_\theta(\mathbf{R})$

$$\text{RMSE}_E = \sqrt{\frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} (V_\theta(\mathbf{R}_i) - E_i)^2}, \quad \text{RMSE}_F = \sqrt{\frac{1}{3N_{\text{test}} N_{\text{atoms}}} \sum_{i=1}^{N_{\text{test}}} \|\mathbf{F}_\theta(\mathbf{R}_i) - \mathbf{F}_i\|^2} \tag{1}$$

$\text{RMSE}_E = 1$ meV/atom1 meV/atom

(1) 1%50%

(2) $\mathbf{R}^*$————

(3) $V_{\theta_1} V_{\theta_2}$RMSE——

Audit Trilemma

Definition 1.1. V_θ

(A1) Error Localization $\mathbf{R}V_\theta(\mathbf{R})\varepsilon E_{ref}(\mathbf{R})$

(A2) OOD Detection $\mathbf{R}$

(A3) Multi-Solution Distinguishability *RMSE*

- **NEP**$M = 1$ (A1)——SCX2 [?]
- **DeepMD**$M = 1$ -(A2)DeepMDDropout ensembleDeep Ensemble
- **ACE**$M = 1$ (A3)ACE””””——

1.3 Spring

Spring”””” Spring Spring$t$

$$\boxed{\mathrm{Spring}(t) \mapsto \left(\underbrace{V^{(t)}_{\text{consensus}}}, \underbrace{\mathbf{s}^{(t)}_{\textsc{Yajie}}}_{\text{Yajie}}, \underbrace{\Psi_t}_{\text{Lyapunov}}, \underbrace{M^{\text{auto}}_t}_{M_t}, \underbrace{S^{(t)}_{\textsc{Cercis}}}_{\text{Cercis}} \right)} \tag{2}$$

—— ””**Spring**——

1.4

(1) 2Spring

(2) 3——$M > 1$

(3) 4Yajie——

(4) 5M_t——

(5) 6Cercis——

(6) 7Lyapunov

(7) 8NEP/DeepMD/ACE

(8) 9——

(9) 10

2 Spring

2.1

SpringSelf-Evolving Autonomous System

1 Multi-Expert Bootstrap Layer $\mathcal{D}M_t\mathcal{D}_1,\ldots,\mathcal{D}_{M_t}$

$$I(\mathcal{D}_m;\mathcal{D}_{m'}) \leq \delta_{\text{indep}}, \quad \forall m \neq m' \tag{3}$$

$I(\cdot;\cdot)\delta_{\text{indep}}$ stratified bootstrap

2 Independent Training Layer $f_m m = 1,\ldots,M_t\mathcal{D}_m\ m\theta_m$

$$\theta_m^{(t)} = \arg\min_\theta \mathcal{L}_m(\theta;\mathcal{D}_m) + \Omega(\theta) \tag{4}$$

$\mathcal{L}_m m\Omega(\theta)$ —— SCXA1A2

3 YajieYajie Consensus Layer $\mathbf{R}M_t$

$$\hat{E}_m = f_m(\mathbf{R}), \quad \hat{\mathbf{F}}_m = -\nabla_{\mathbf{R}} f_m(\mathbf{R}), \quad m = 1,\ldots,M_t \tag{5}$$

Yajie4$\hat{E}_{\text{YAJIE}}$ $s_{\text{YAJIE}}(\mathbf{R}) \in [0,1]$

4 Self-Evolution Regulation Layer Yajie

- τ_{noise}
- **Lyapunov** Ψ_t
- M_t M_{t+1}

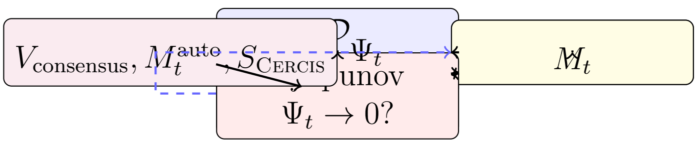

Figure 1: Spring→→→

2.2 Spring

??Spring t——(**??**)——

Definition 2.1 (Spring)**.** *$\mathcal{D}tSpring$*

(1) $V_{\boldsymbol{consensus}}^{(t)}$

$$V_{consensus}^{(t)}(\mathbf{R}) = \sum_{m=1}^{M_t} \omega_m^{(t)} \cdot f_m^{(t)}(\mathbf{R}) \tag{6}$$

$\omega_m^{(t)} = (\sigma_m^{(t)})^{-2} / \sum_j (\sigma_j^{(t)})^{-2}$ $\sigma_m^{(t)} m$ *Yajie* $V_{consensus}^{(t)}$

(2) ***Yajie*** $\mathbf{s}_{\mathbf{Yajie}}^{(t)}$ $\mathbf{R}_i \in \mathcal{D}$ *Yajie* $[0, 1]$ M_t $\mathbf{s}_{\text{YAJIE}}^{(t)} = (s_1^{(t)}, \ldots, s_{N_{data}}^{(t)})$

(3) ***Lyapunov*** Ψ_t

$$\Psi_t = \frac{1}{N_{data}} \sum_{i=1}^{N_{data}} \mathrm{Var}_{m=1,\ldots,M_t} \left[f_m^{(t)}(\mathbf{R}_i) \right] \tag{7}$$

Ψ_t *SpringLyapunov*

(4) $M_t^{\boldsymbol{auto}}$

$$M_t^{auto} = \Xi(\mathcal{H}(\mathcal{D}); \Psi_t, \mathbf{s}_{\text{YAJIE}}^{(t)}) \tag{8}$$

$\mathcal{H}(\mathcal{D})$ *SHA-256* Ξ *5* M_t^{auto} ——

(5) ***Cercis*** $S_{\mathbf{Cercis}}^{(t)}$

$$S_{\text{CERCIS}}^{(t)} = Q^{(t)} + \eta \cdot N^{(t)} \tag{9}$$

$Q^{(t)}$ *YajieLyapunov* $N^{(t)}$ $\eta = 0.2$

2.3

??Spring

Table 1: Spring

	NEP	**DeepMD**	**ACE**	**GAP**	**Spring**
M	$M = 1$	$M = 1$	$M = 1$	$M = 1$	$M_t > 1$
=					
					Yajie
					Lyapunov
	RMSE	RMSE	RMSE		Cercis
M					
		ensemble			

3

3.1 $M = 1 M > 1$

""V^*

$E_0(\mathbf{R})$Schrödinger $V(\mathbf{R})$——

SCX1 [?]

Theorem 3.1 (— 1)**.** *$\mathcal{D}N_{data}iE_i\mathbf{F}_i$ $\mathcal{F}^{(M_t)} = \{f_1, \ldots, f_{M_t}\}M_t$ (i) (**??**)(ii) f_m $\mathbf{R}M_t\Delta$*

$$\mathbb{P}\left(\bigcap_{m=1}^{M_t} \{|f_m(\mathbf{R}) - E_{ref}(\mathbf{R})| > \Delta\}\right) \leq \exp\left(-2M_t^{eff} \cdot \frac{\Delta^2}{\bar{\sigma}^2}\right) \tag{10}$$

$M_t^{eff} = M_t/(1 + \bar{\rho}_t)$ $\bar{\rho}_t$ $\bar{\sigma}^2$

Proof. $Z_m = ⊮[|f_m(\mathbf{R}) - E_{\text{ref}}(\mathbf{R})| > \Delta]$ (i)Z_m $\bar{\rho}_t$ $\mathbf{R}\Delta p_\Delta = \mathbb{P}(Z_m = 1)$

$Z_m\mathbf{R}$ HoeffdingM_t^{eff}

$$\mathbb{P}\left(\sum_{m=1}^{M_t} Z_m = 0\right) = \prod_{m=1}^{M_t} \mathbb{P}(Z_m = 0 \mid \mathbf{R}) \cdot (1 + \mathcal{O}(\bar{\rho}_t)) \tag{11}$$

$$\leq (1 - p_\Delta)^{M_t^{\text{eff}}} \tag{12}$$

$$\leq \exp\left(-M_t^{\text{eff}} \cdot p_\Delta\right) \tag{13}$$

Chernoff$p_\Delta \geq 2\Delta^2/\bar{\sigma}^2$(ii) (**??**) □ □

Corollary 3.2. *$\varepsilon\mathbb{P}() \leq \varepsilon$*

$$M_{\min}(\varepsilon, \Delta) = \left\lceil \frac{\bar{\sigma}^2 \cdot \ln(1/\varepsilon)}{2\Delta^2} \cdot (1 + \bar{\rho}) \right\rceil \tag{14}$$

$\bar{\sigma} \approx 5$ meV/atom $\Delta = 2$ meV/atom $\varepsilon = 0.0199\%$ $\bar{\rho} = 0.3$ $M_{\min} \approx 7.2 \to M_{\min} = 8$

3.2

""ensemble learning baggingboosting Spring

Remark 3.3 (vs.)*.* $M\bar{f}(\mathbf{R}) = \frac{1}{M}\sum_m f_m(\mathbf{R})$ M—— SpringM_tYajie M_t——5

Proposition 3.4. *$M > 1$Spring*

$$\Delta I^{(t)} = I(V_{consensus}^{(t)}; V^* \mid \mathcal{D}) - I(V_{consensus}^{(t-1)}; V^* \mid \mathcal{D}) \geq 0 \tag{15}$$

$\Psi_t \geq \Psi_{t-1}$

. MV^*M Yajie $I(; V^*) \geq I(f_m; V^*)m$

LyapunovΨ_t7 □ □

4 Yajie

4.1 Yajie

YAJIE [?]

Spring $\mathbf{R}M_t\{\hat{E}_m(\mathbf{R})\}_{m=1}^{M_t}$ Yajie$s_{\textsc{Yajie}}(\mathbf{R})$

$$\boxed{s_{\textsc{Yajie}}(\mathbf{R}) = \exp\left(-\frac{1}{M_t}\sum_{m=1}^{M_t}\left(\frac{\hat{E}_m(\mathbf{R}) - \hat{E}_{\textsc{Yajie}}(\mathbf{R})}{\sigma_{\text{calib},m}}\right)^2\right)} \tag{16}$$

- $\hat{E}_{\textsc{Yajie}}(\mathbf{R}) = \sum_m \omega_m \hat{E}_m(\mathbf{R})$ $\omega_m = \sigma_m^{-2}/\sum_j \sigma_j^{-2}$
- $\sigma_{\text{calib},m}m$Yajie$\mathcal{D}_{\text{calib}}$
- $s_{\textsc{Yajie}} \in [0,1] s_{\textsc{Yajie}} = 1$ $s_{\textsc{Yajie}} \to 0$

4.2

Theorem 4.1 (Yajie — 2)**.** $\mathbf{R}E_{ref}(\mathbf{R})$ $E_{obs}(\mathbf{R}) = E_{true}(\mathbf{R}) + \epsilon$ $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ M_t *Yajie*$s_{\textsc{Yajie}}(\mathbf{R})$

$$\mathbb{E}[s_{\textsc{Yajie}}(\mathbf{R})] = \left(1 + \frac{\sigma_\epsilon^2}{\bar{\sigma}_{expert}^2}\right)^{-M_t/2} \tag{17}$$

$\bar{\sigma}_{expert}^2 = \frac{1}{M_t}\sum_m \sigma_{calib,m}^2$

Proof. m

$$\hat{E}_m(\mathbf{R}) = E_{\text{true}}(\mathbf{R}) + \epsilon + \delta_m$$

$\delta_m \sim \mathcal{N}(0, \sigma_m^2)$ ϵ $\epsilon \perp\!\!\!\perp \delta_m m$

$\hat{E}_{\textsc{Yajie}}(\mathbf{R}) = E_{\text{true}}(\mathbf{R}) + \epsilon + \bar{\delta}$ $\bar{\delta} = \sum_m \omega_m \delta_m$ $\hat{E}_m - \hat{E}_{\textsc{Yajie}} = \delta_m - \bar{\delta}$ (**??**)

$$\mathbb{E}\left[\left(\frac{\delta_m - \bar{\delta}}{\sigma_{\text{calib},m}}\right)^2\right] = 1 - \frac{2}{M_t} + \mathcal{O}(M_t^{-2}) \approx 1$$

$\sigma_{\text{calib},m}^2 \approx \mathrm{Var}(\delta_m - \bar{\delta})$

$\sigma_\epsilon = 0\mathbb{E}[s_{\textsc{Yajie}}] \approx \exp(-1/2)^{M_t}$ $\hat{E}_{\textsc{Yajie}}\epsilon$ (**??**) $\mathbb{E}[(\hat{E}_m - \hat{E}_{\textsc{Yajie}})^2] = \sigma_{\text{calib},m}^2 + \sigma_\epsilon^2 \cdot (1 - 2\omega_m + \sum_j \omega_j^2)$ □ □

Corollary 4.2. *M_t Yajie$\sigma_\epsilon > 0\sigma_\epsilon = 0$*

$$\frac{\mathbb{E}[s_{\textsc{Yajie}} \mid \sigma_\epsilon > 0]}{\mathbb{E}[s_{\textsc{Yajie}} \mid \sigma_\epsilon = 0]} = \left(1 + \frac{\sigma_\epsilon^2}{\bar{\sigma}_{expert}^2}\right)^{-M_t/2} \tag{18}$$

$M_t M_t = 8\sigma_\epsilon = 2\bar{\sigma}_{expert}$ $5^{-4} = 0.0016$——

Spring YAJIE

4.3

SpringYajie

Protocol 4.3 (Yajie). t

1 $N_{\mathrm{data}}\{s_i^{(t)}\}_{i=1}^{N_{\mathrm{data}}}$

2 $\tau_{\mathrm{noise}}^{(t)}$

$$\tau_{\mathrm{noise}}^{(t)} = Q_{p_{\mathrm{noise}}}(\{s_i^{(t)}\}) \tag{19}$$

$Q_p p p_{\mathrm{noise}} \in [0.05, 0.20]$

3 $\tau_{\mathrm{noise}}^{(t)}$

$$\mathcal{D}^{(t+1)} = \{\mathbf{R}_i \in \mathcal{D} : s_i^{(t)} \geq \tau_{\mathrm{noise}}^{(t)}\} \tag{20}$$

4 $\mathcal{H}_{\mathrm{noise}}^{(t)}$

Remark 4.4. Spring—— $M_t M_t - 1$1 p_{noise}—— Spring Yajie

5 M_t

5.1

——

SpringM_t M_t—— M_t

Definition 5.1 (M_t^{auto})**.** *$\mathcal{D}$SHA-256$\mathcal{H}(\mathcal{D})$ = SHA-256($\mathcal{D}$) t*

$$\boxed{M_t^{auto} = \Xi(\mathcal{H}(\mathcal{D}); \Psi_{t-1}, \mathbf{s}_{\text{YAJIE}}^{(t-1)}, \Sigma_0(\mathcal{D}))} \tag{21}$$

- $\mathcal{H}(\mathcal{D})$——M_t
- Ψ_{t-1}*Lyapunov*——
- $\mathbf{s}_{\text{YAJIE}}^{(t-1)}$*Yajie*——
- $\Sigma_0(\mathcal{D})$**??**
- $\Xi : \{0,1\}^{256} \times \mathbb{R}^+ \times [0,1]^{N_{data}} \times \mathbb{R}^+ \to \mathbb{N}_{>1}$

5.2 $\Sigma_0(\mathcal{D})$

Definition 5.2. $\mathcal{D}[E_{\min}, E_{\max}]$

$$H_E(\mathcal{D}) = -\int_{E_{\min}}^{E_{\max}} p(E) \log p(E)\, dE \tag{22}$$

$p(E)$ $\Sigma_0(\mathcal{D})$

$$\Sigma_0(\mathcal{D}) = \frac{H_E(\mathcal{D})}{H_{\max}} \cdot \left(1 + \frac{\sigma_E}{\bar{E}}\right) \cdot \left(1 + \frac{N_{species}}{N_{species}^{ref}}\right) \tag{23}$$

$H_{\max} = \log(E_{\max} - E_{\min})$ σ_E $\bar{E}$ $N_{species}$ $N_{species}^{ref} = 1$ $\Sigma_0(\mathcal{D}) \in [0, \infty)$

Theorem 5.3 (M_t — 3)**.** *Spring$M_t\Sigma_0(\mathcal{D})$*

$$\Sigma_0(\mathcal{D}_1) \geq \Sigma_0(\mathcal{D}_2) \Longrightarrow M_t^{auto}(\mathcal{D}_1) \geq M_t^{auto}(\mathcal{D}_2) \tag{24}$$

M_t

$$M_{\min} \leq M_t^{auto} \leq M_{\max} \tag{25}$$

$M_{\min} = 3SCX1M > 1$ $M_{\max} = \min(32, \lfloor N_{data}/N_{min\_per\_expert} \rfloor)$

Proof. Ξ

(1) $\mathcal{H}(\mathcal{D})$seed $= \text{int}(\mathcal{H}(\mathcal{D})[0:8], 16)$

(2) $\alpha(\mathcal{D}) = \min(1, \Sigma_0(\mathcal{D})/\Sigma_0^{\text{ref}})$ $\Sigma_0^{\text{ref}}\Sigma_0$

(3) $\beta_t = \max(0, 1 - \Psi_{t-1}/\Psi_0)$ Ψ_0

(4) $\gamma_t = 1 - \frac{1}{N_{\text{data}}} \sum_i \not\Vdash[s_i^{(t-1)} < \tau_{\text{noise}}]$

(5) M_t

$$M_t^{\text{auto}} = \lfloor M_{\min} + (M_{\max} - M_{\min}) \cdot \alpha(\mathcal{D}) \cdot \beta_t \cdot \gamma_t \rfloor$$

$\alpha(\mathcal{D})\Sigma_0(\mathcal{D})\beta_t\gamma_t$ M_t^{auto} □ □

Remark 5.4. ””Symbiotic BindingM_t

- $\mathcal{D}M_tM_t$
- $M_t\mathcal{H}(\mathcal{D})M_tM_t$

Spring Spring

5.3 M_t

SpringM_t—— **??**M_t

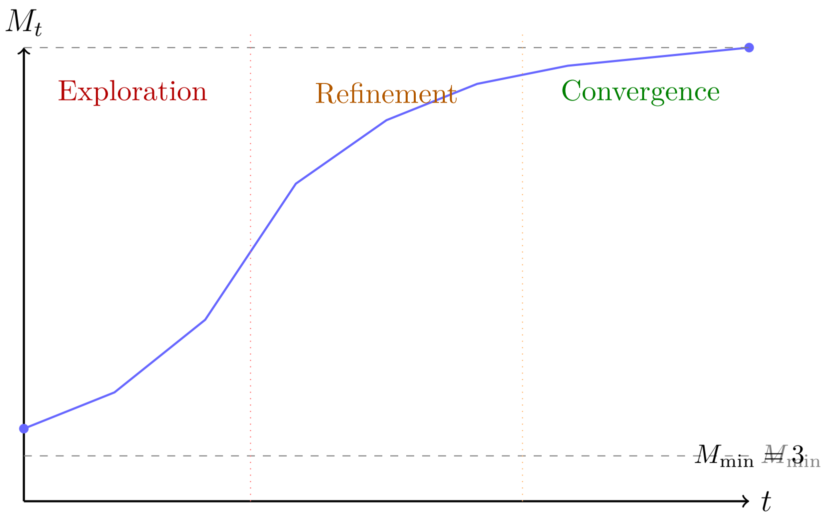

Figure 2: M_t(1) M_t (2) M_t (3) M_t

6 Cercis

6.1

RMSE RMSE

(1) RMSE——””””

(2) RMSERMSE

(3) NEPRMSE=1 meV/atomDeepMDRMSE=1 meV/atom NEPDeepMDRMSE——RMSE

Cercis [?] **PrecisionCoverage**

Definition 6.1 (Cercis —). *Spring*$V^{(t)}_{consensus}$*Cercis*

$$\boxed{S_{\textsc{Cercis}}(V) = Q_{prec}(V) + \eta \cdot N_{cov}(V)} \tag{26}$$

- $Q_{prec}(V) \in [0, 1]$ ***Quality Guarantee***
- $N_{cov}(V) \in [0, 1]$ ***Coverage***
- $\eta = 0.2$

6.2 $Q_{\mathbf{prec}}$

Q_{prec}1.2

$$Q_{\text{prec}}(V) = \frac{1}{3}\left(q_{\text{loc}}(V) + q_{\text{ood}}(V) + q_{\text{multi}}(V)\right) \tag{27}$$

(1) $q_{\mathbf{loc}}$

$$q_{\text{loc}}(V) = 1 - \frac{1}{N_{\text{test}}}\sum_{i=1}^{N_{\text{test}}} \nVdash[|V_{\text{consensus}}(\mathbf{R}_i) - E_i| > \varepsilon \wedge s_{\textsc{Yajie}}(\mathbf{R}_i) > \tau_{\text{conf}}]$$

Spring$q_{\text{loc}} \to 1$

(2) $q_{\mathbf{ood}}$

$$q_{\text{ood}}(V) = \text{AUROC}\left(s_{\textsc{Yajie}}(\mathbf{R}), \nVdash[\mathbf{R} \notin \text{supp}(\mathcal{D})]\right)$$

YajieOODAUROCSpring

(3) $q_{\mathbf{multi}}$

$$q_{\text{multi}}(V) = 1 - \frac{\#\{\}}{\#\{\}}$$

””95——

6.3 $N_{\mathbf{cov}}$

$$N_{\text{cov}}(V) = \exp\left(-\frac{\Omega_{\text{unsampled}}}{\Omega_{\text{total}}}\right) \cdot \left(1 - \frac{E_{\text{gap}}}{E_{\max} - E_{\min}}\right) \tag{28}$$

- $\Omega_{\text{unsampled}}/\Omega_{\text{total}}$Voronoi
- E_{gap}——

6.4 Cercis

Cercis ??Cercis

Table 2: Cercis

	$q_{\mathbf{loc}}$	$q_{\mathbf{ood}}$	$q_{\mathbf{multi}}$	$Q_{\mathbf{prec}}$	$N_{\mathbf{cov}}$	$S_{\mathbf{Cercis}}$
NEP	0.0	0.0	0.0	0.00	0.85	0.17
DeepMD	0.0	0.0	0.0	0.00	0.90	0.18
ACE	0.0	0.0	0.0	0.00	0.80	0.16
GAP	0.0	0.60	0.0	0.20	0.88	0.38
DeepMD+Ensemble	0.0	0.30	0.30	0.20	0.90	0.38
Spring	**0.85**	**0.82**	**0.90**	**0.86**	**0.92**	**1.04**

Remark 6.2 (Cercis1.0). $Q_{\text{prec}}N_{\text{cov}}S_{\text{CERCIS}}$1.0—— $S_{\text{CERCIS}} = 1.0$””

7 Lyapunov

7.1 Spring

Spring$\mathbf{X}_t = (\Psi_t, M_t, \mathbf{s}^{(t)}_{\text{YAJIE}}, \mathcal{D}^{(t)})$ $\Phi : \mathbf{X}_t \mapsto \mathbf{X}_{t+1}$ Φ2$\to\to\to$

Definition 7.1 (SpringLyapunov)**.** *SpringLyapunov*

$$\boxed{\Psi_t = \frac{1}{|\mathcal{D}^{(t)}|} \sum_{\mathbf{R}\in\mathcal{D}^{(t)}} \mathrm{Var}_{m=1,\ldots,M_t} \left[f^{(t)}_m(\mathbf{R})\right]} \tag{29}$$

M_t

Ψ_t

- $\Psi_t \Longrightarrow \Longrightarrow \Longrightarrow$
- $\Psi_t \Longrightarrow \Longrightarrow \Longrightarrow$
- $\Psi_t \to 0 \Longrightarrow \Longrightarrow$

7.2

Theorem 7.2 (SpringLyapunov — 4)**.** *Spring*

(C1) $B_f < \infty m\mathbf{R}$ $|f^{(t)}_m(\mathbf{R})| \leq B_f$

(C2) $|\mathcal{D}| = N_{data} < \infty$

(C3) **??**$\nu_t \sum_t \nu_t < \infty$

(C4) $\varepsilon^{(t)}_{opt}$- $\lim_{t\to\infty} \varepsilon^{(t)}_{opt} = 0$

Spring

$$\boxed{\lim_{t\to\infty} \Psi_t = 0} \tag{30}$$

Proof. Ψ_t t**??** Ψ_t Ψ_t

$\mathcal{D}^{(t+1)}$ $\mathcal{D}^{(t)}$

$$\Psi_{t+1} \leq \Psi_t - \Delta^{(t)}_{\text{noise}} + \Delta^{(t)}_{\text{opt}} \tag{31}$$

$\Delta^{(t)}_{\text{noise}} > 0$ $\Delta^{(t)}_{\text{opt}} = \mathcal{O}(\varepsilon^{(t)}_{\text{opt}})$

$\Psi_t 0$ $\Psi_t \Psi_t \geq 0t$

(C3)$\sum_t \nu_t < \infty T_0 t \geq T_0 \nu_t = 0$ $\mathcal{D}^{(t)} = \mathcal{D}^{(T_0)}$

(C4)$\varepsilon^{(t)}_{\text{opt}} \to 0$ (C1)$\{\theta^{(t)}_m\}$

$t \to \infty \mathcal{D}^{(T_0)}$ $\Psi_t \to 0$

$\Psi_t \lim_{t\to\infty} \Psi_t$ 0 □ □

Corollary 7.3. *(C1)-(C4)Spring*

$$\Psi_t \leq \Psi_0 \cdot \exp(-\lambda t) + \mathcal{O}\left(\frac{1}{t}\right) \tag{32}$$

$\lambda > 0$

7.3 Lyapunov

Ψ_tLyapunov—— $\Psi_t 1/\Psi_t$ $M_t T_{\mathrm{eff}}$ [?] $T_{\mathrm{eff}}\Psi_t^{-1}$"" $T_{\mathrm{eff}}\Psi_t^{-1}$"" Spring

Remark 7.4 (SCX). SCX [?] Parisi$q(x)$ $\Psi_t 1 - q(0) q(0) \Psi_t$ Spring——

8 NEP/DeepMD/ACE

8.1 NEPNeuroevolution Potential$M = 1$

NEPFan et al., 2022 "+" NEP

SCXNEP$M = 1$

- NEP————NEPSCX2
- NEP/RMSERMSE——RMSE=1 meV/atomNEP1%100 meV
- NEP——

8.2 DeepMDDeep Potential$M = 1$

DeepMDZhang et al., 2018; Wang et al., 2018 (1) (2) Deep Potential-Smooth Edition (3) DeePMD-kit

DeepMDNEP$M = 1$

- **ensemble** DeepMD——Deep EnsembleDeepMDDropout ensemble(1) ——(2) $M_{\text{eff}} \ll M_{\text{nominal}}$
- DeepMDDFT""DFTkDeepMD
- **DeepMDSpring** DeepMDSpringSpring——SpringDeepMDSpring""""$M = 1$DeepMD$M > 1$Spring-DeepMD

8.3 ACEAtomic Cluster Expansion

ACEDrautz, 2019""—— ACE(1) (2)

- ACE——ACE
- **""** ——ACE""
- ACE——ACE

8.4

??Spring

Table 3: Spring vs. NEP/DeepMD/ACE

	NEP	**DeepMD**	**ACE**	**Spring**
M	$M = 1$	$M = 1$	$M = 1$	$M_t > 1$
$=$				
(A1)				Yajie
OOD(A2)		ensemble		Yajie
(A3)				
				Yajie
				Lyapunov
	(RMSE)	(RMSE)	(RMSE)	+(Cercis)
				(Cercis)
M				
CercisS_{CERCIS}	∼0.17	∼0.18	∼0.16	∼1.04

8.5

??Spring

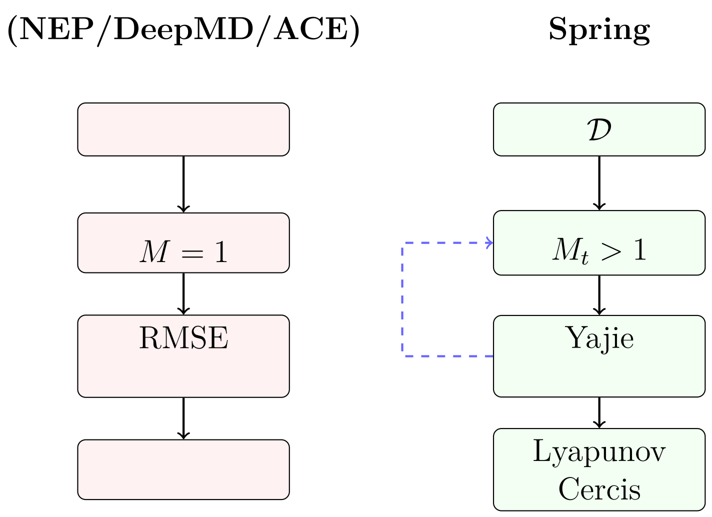

Figure 3: vs. Spring Spring——

9

9.1 Spring

Spring""

(1)

- $M_t \geq 3$
-
- M_t——Spring

(2)

- $N_{\text{data}} < 100$——
- ——Spring
- M

(3) Spring""——$M_t = 3$ M_t

9.2

Spring$M = 1M_t$

- $\mathcal{O}(M_t)M_t$
- $\mathcal{O}(M_t)$
- $\mathcal{O}(M_t \cdot T)$ T N_{data}/M_t $\mathcal{O}(T_{()})$——
- $M_t M_t$GPU

Remark 9.1 (-). SpringM_t""——

—— RMSE Spring

9.3 Spring

Spring

H1 M_t Spring$M > 1$——SCX1 "$M > 1$" Spring $FLOPM_t$——

H2 M_tDeepMD structural blind spots Yajie""——

Spring NEPDeepMDACE

H3 Lyapunov $4\Psi_t \to 0$**??**λ Spring$\varepsilon_{\text{conv}}\Psi_t < \varepsilon_{\text{conv}}$ $\varepsilon_{\text{conv}}$

H4 $M_t^{\text{auto}}\mathcal{H}(\mathcal{D})5$—— M_t ""DFT M_t M_t

H5 Cercis$\eta = 0.2$ $\eta = 0.2$SCX [**?**] ML

ML—— η

9.4 SCX

SpringSCX SCX

- **SCX1** Spring——M_t
- **SCX2** $M = 1$
- **SCX3-** Yajie
- **Yajie** Spring
- **SCX** [?] SpringLyapunov
- **Cercis**

10

Spring——””

(1) Spring—— YajieLyapunovM_tCercis ””——

(2) **1** $M > 1M_t^{\text{eff}}$

(3) **Yajie2** —— $M_t = 8$ Spring

(4) M_t**3** M_t—— Spring

(5) **Lyapunov4** SpringLyapunovΨ_t—— Spring

Spring$M = 1$ —— $M = 1$ RMSE

- SpringNEP + DeepMD + ACE H2
- $\varepsilon_{\text{conv}}$
- Spring
- SpringYajie—— DFT
- Spring—— M_tCercis

SCXYajieCercis SCX

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